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Sensitivity Analysis of a TIR-Based Cold-Water Patch Detection Algorithm

The Influence of Cold Water Tributaries on Cold Water Patches

Bachelor Thesis

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1 Theoretical Background

1.1 Model - Working Definition

The term model is widely used across environmental and earth sciences, yet a precise definition of a model is surprisingly challenging to find in literature. In this work, a model is understood as a set of simplified laws and processes, intertwined by dependencies, that together form a reduced reflection of an underlying system. The aim is to capture the essential behavior of that system and enable analysis from which inferences about reality can be drawn. This view is consistent with Saltelli et al. (2008), who, drawing on Rosen (1991), describe the modelled portion of reality as an arbitrary enclosure of an otherwise open and interconnected system.

1.2 Model Input Sensitivity Analysis

The inputs to models can diver and are generally dependent in their structure and manyfoldness on the nature of the model itself. Under the light of sensitivity analysis however, it can be worthwhile to define, what the inputs to a model actually contains. In sensitivity analysis, everything that causes variation in the output of the model is classified as an input (Satelli_Marco). Providing an example. If a linear regression of a target variable Y is performed, then not only are the predictors $X_1, \dots, X_n$ considered as model inputs, but also the parameters $\theta_1, \dots, \theta_n$ that correspond to the predictors.

1.3 Sensitivity Analysis

According to Saltelli et al. sensitivity analysis is "The study of how uncertainty in the output of a model [. . .] can be apportioned to different sources of uncertainty in the model input" (Saltelli et al., 2004). Sensitivity analysis provides a range of benefits. Firstly it allows the Modeler to apply a strategy called "Factor Fixing". If we discover, that the model output Y is independent of one or multiple parameters θ_i, then we can set these to fixed constants. It may then even be possible to condense a complete submodule of a model, if all parameters connected to this sub-module are noninfluential. So sensitivity analysis can be a tool for model simplification (Saltelli_Marco). Secondly, sensitivity analysis generally improves model validity by identifying highly influential and therefore critical model components (Ligmann-Zielinska). A range of different methods for sensitivity analysis exist. They can be grouped into global and local analysis methods.

Local methods compute a sensitivity score for each parameter at a single point in the input space, typically the baseline point, which is customarily the best estimate point for the parameters. Scores are based on partial derivatives and are therefore only informative for linear models, where the derivative at one point represents behaviour across the entire input space (Saltelli and Annoni 2010).

Global sensitivity analysis methods explore the entire input space by varying all input parameters simultaneously across their plausible ranges. Sensitivity scores are therefore not tied to a single point but reflect average behaviour across the full parameter space. This makes global methods valid regardless of model linearity, as no assumption about model structure is required. (Saltelli and Annoni 2010).

global methods

1.3.1 The Morris Method

The Morris Method belongs to the group of screening methods in sensitivity analysis. These are ideal to filter a short list of important input parameters from computationally expensive models, with a great number of input parameters. This can be done, as these methods only require a relatively little number of model evaluations to provide sensitivity measures for the input parameters. This makes these methods a good choice when the target of the sensitivity analysis is Factor Fixing. (Satelly and Tarantola). The following explenation of morris method closely follows Saltelli_Marco and is explained more explicitly in their chapter.

At the core of this introduction sits a k-dimensional model Y with inputs $X_1, \ldots, X_k$. These inputs all together can also be written in a vector $\mathbf{X}$. The model inputs are defined in the k-dimensional unit cube, meaning that X_i can take on a value in the interval $[0, 1]$ (Saltelli, Tarantola). It is worth pointing out, that every model with a continuous input can be scaled to the $[0, 1]$ range, which makes the unit cube a great place to define methods (Claude). In the Morris method, this input space is discretized into a grid Ω. This grid features p levels.

An elementary effect of the ith input variable is then defined as:

$$EE_i = \frac{Y(X_1, X_2, \ldots, X_{i-1}, X_i + \Delta, \ldots, X_k) - Y(X_1, X_2, \ldots, X_k)}{\Delta} \tag{1-1}$$

Where Δ represents some step in the unit cube. Δ can be chosen from a range of different values, which are given by the set $\left\{\frac{1}{p-1}, \ldots, 1 - \frac{1}{p-1}\right\}$. Which value to choose for Δ from this set has implications and will be addressed in a later paragraph.

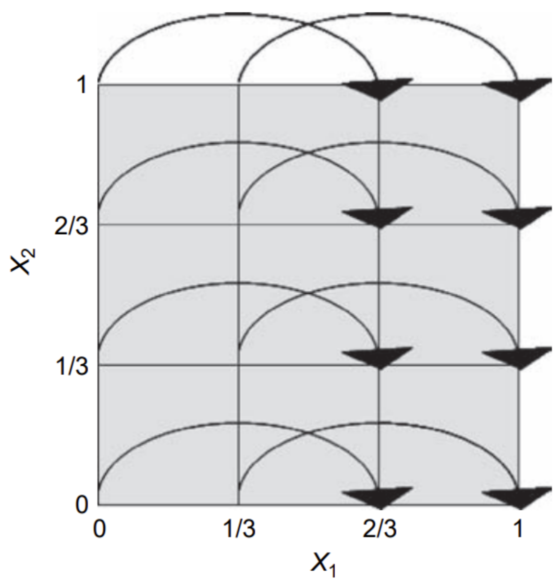

Abb. 1-1: An example grid Ω. The grid features two parameters X_1, X_2, meaning that it is a two-dimensional grid. Further, it features four levels ($p = 4$). The step length Δ chosen here is $\frac{2}{3}$, which corresponds to an element of the set of viable values for Δ: $1 - \frac{1}{4-1} = 1 - \frac{1}{3} = \frac{2}{3}$.

In the definition of the elementary effect, the vector $\mathbf{X} = (X_1, X_2, \ldots, X_k)$ has to fulfil the condition that $\mathbf{X} + \mathbf{e}_i \cdot \Delta$ still has to lie within the grid Ω. $\mathbf{e}_i$ is a vector of zeros except having a one at its ith entry. So for instance:

$$\mathbf{e}_2 = (0, 1, 0, \ldots, 0) \tag{1-2}$$

The Morris method occupies itself with finding the statistical distribution of each of the i elementary effects. $\mathcal{F}_i$ is thereby used to denote the distribution of EE_i, in other words $\mathcal{F}_i \sim EE_i$. $\mathcal{F}_i$ is a finite discrete distribution, and its number of elements depends on the number of grid levels p, which is however not that much of interest in an applied setting.

How are the EE_i sampled? Generally many different points $\mathbf{X}$ over the whole grid are sampled and then used to compute elementary effects EE_i. This guarantees, that each elementary effect EE_i is computed for many areas within the grid. This is also the reason, why the Morris method is generally classified as a global method. The exact sampling scheme is out of this project's scope but is covered in detail in Saltelli_Marco chapter 3.3.

Once the distributions $\mathcal{F}_i$ have been mapped out, statistical measures can be computed just like over every other statistical distribution. In Morris, the mean μ_i and the standard deviation σ_i of $\mathcal{F}_i$ are computed. These two statistical measures together already allow for reasonable insight into the influence of the ith parameter X_i which corresponds to the ith elementary effect EE_i:

If μ_i deviates strongly from 0, then this means that the elementary effect EE_i generally takes on non-zero values and therefore that the model output changes notably when a step of Δ is performed along the dimension spanned by X_i. This then ultimately means, that changes in X_i seem to generally affect the model output Y considerably and X_i therefore receives a high sensitivity score. Generally, the more μ_i deviates from zero, the larger the sensitivity of the model output towards X_i.

If σ_i is large, this means that the distribution $\mathcal{F}_i$ is spread out considerably over a larger range. This indicates, that the elementary effects EE_i that were computed were generally different depending on their location in the unit cube. Meaning that in one area, the model might only yield small changes in output, when X_i is varied while in other areas X_i leads to strong changes in model output. This behaviour points either to nonlinearity regarding X_i or to the fact that X_i interacts with other input parameters. Which effect is responsible for the high standard deviation (or whether both are responsible for the effect at the same time) can not be determined. Which is a shortcoming of the method of Morris. Figure 1-2 provides an visual example of possible F distributions of a Factor X_i with the associated μ and σ values and the corresponding interpretation.

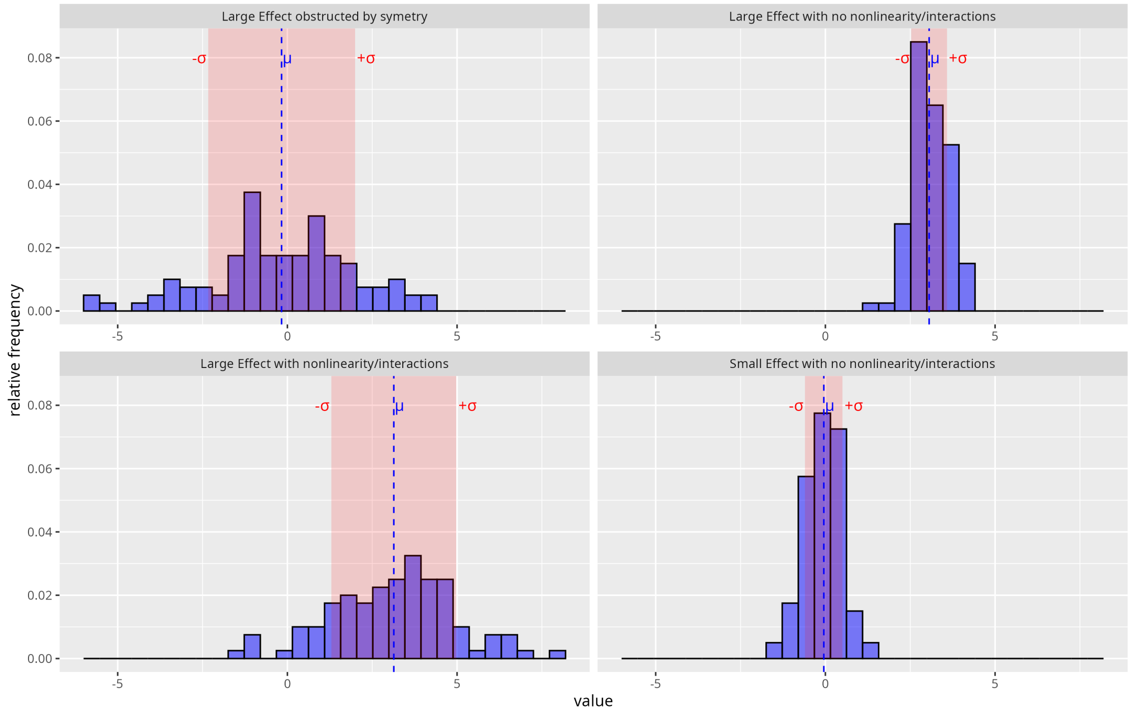

Abb. 1-2: Different F distributions of a parameter X_i are shown. Additionally, μ and σ of each distribution is provided. An interpetation is provided

Situations can occur, where $\mathcal{F}_i$ contains many extreme values however the distribution is symmetric around zero. Then computing the mean value μ_i will yield a value close to zero and we would assume, that the parameter X_i therefore is noninfluential. This would be a faulty assumption, as the sampled elementary effects EE_i are substantial in their magnitude but just cancel themselves out. This phenomenon can be spotted by always taking σ_i into consideration, as in this scenario σ_i is large, but it is more common to also sample an additional distribution $\mathcal{G}_i$ which captures the magnitude of the elementary effects $|EE_i|$. So $\mathcal{G}_i \sim |EE_i|$. We then compute the mean of this distribution μ_i^* which can be interpreted like μ_i. The further away from zero, the more influential is the parameter. But since it is the magnitude, the distribution never features negative values and therefore cancellation effects can not occur.

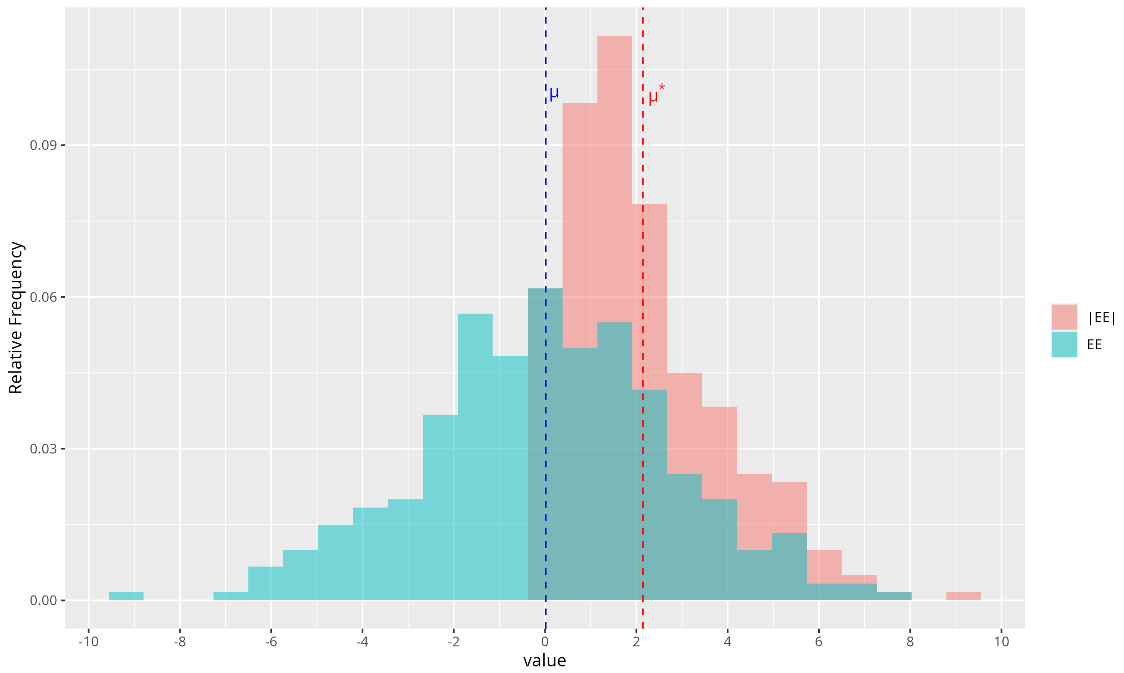

Abb. 1-3: The difference of μ and c visualized with their underlying distribution $\mathcal{G}$ and $\mathcal{F}$. While the mean μ of the distributuion $\mathcal{F}$ lies at around zero, $\mu^{\star}$ significantly deviates from zero

1.4 Coldwater Patches

Cold water patches are areas in rivers and streams that feature a colder water temperature than the temperature of the "hauptgerinne". The temperature difference varies in literature but is typically set to be between 0.5 and 3 degrees Celsius below the "hauptgerinne" temperatur, but this varies over the literature (z.B. Dugdale et al., 2013, 2015; Wawrzyniak et al., 2016; Casas-Mulet et al., 2020). Cold water patches are not automatically a cold water refugia. These are coldwater patches which are also used by coldwater organisms for thermal relief (Meja et all.). The importance of cold water refugia is expected to rise with the effects of climate change (IPCC, 2014)

2 Methods

Figure xy provides a high level overwiev of the methodology deployed to answer the research questions. The mehtod can be split up into two sequential segments, where segment 1 is setting the stage for segment 2. First morris experiment is set up to perform a sensitivity analysis by only varying the buffer width and the step length. The resulting parameter tuples are then deployed in model evaluation, each tuple beeing used to create one set of CWP. The resulting CWP are then used to create fuzzy representations, which allow for the visual determination which CWP are subject to change between parameter tupples. CWP that are subject to change and can be clearly be attributed to either beeing tributary cause or not tributary caused, are then

2.1 Morris Experiment for pixel buffer and step length parameters

2.2 Model Evaluation along Parameter Combinations

2.3 Creation of a fuzzy overlay of CWP

2.4 Selection of CWP

2.5 Morris Experiment for pixel buffer, step length and temperature delta

2.6 Model Evaluation along Parameter Combinations

2.7 Compute total area statistics for selected CWP

2.8 Compute morris scores for pixel buffer, step length and temperature delta

2.9 Usage of KI

Claude Opus 4.6 was used as an aid to ensure proper formulation, spelling and grammatical correctness. Further it was used to derive the LaTeX for Images and simulation protocol table.